



➞ System configurations and diagrams

Memory and storage configurations

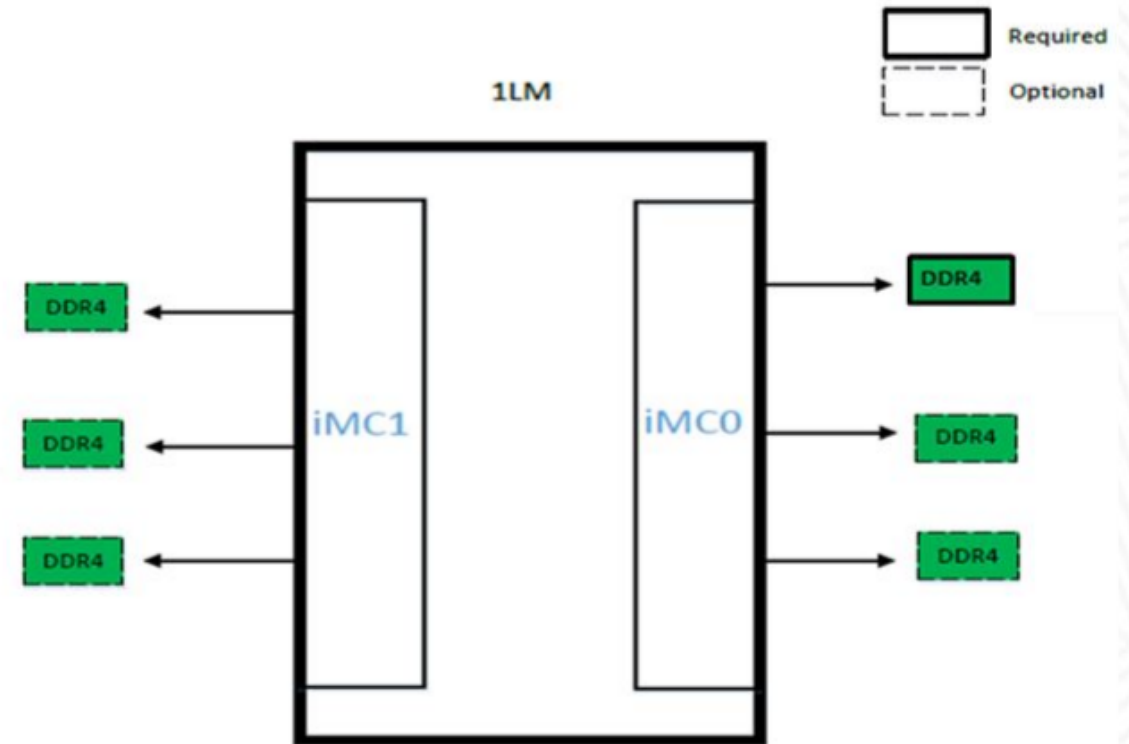


Memory configurations

SR550 supports many DDR4 DIMM types, including Registered DIMMs (RDIMM), Load Reduced DIMMs (LRDIMM) and 3DS RDIMM (available later).

Each processor contains two integrated memory controllers (IMC) with each controller providing three channels of DDR4; one DIMM per channel.

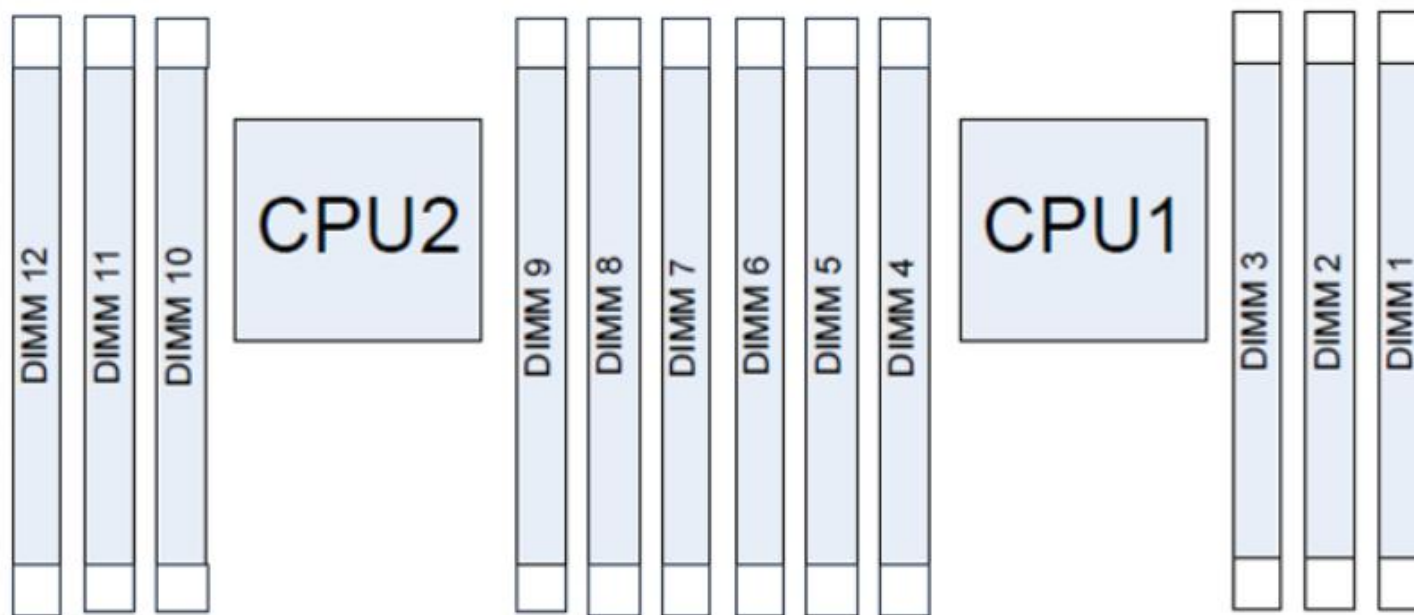
The minimum memory configuration is one DDR4 DIMM per processor.



Note: Even though the system will function with one DIMM, six DIMMs per processor is recommended to improve system performance.

Memory population rules

There are total of 12 memory slots on the system board.



Click each memory mode to view population rules.

Independent
One CPU

Independent
Two CPU

Rank sparing
One CPU

Rank sparing
Two CPU

Mirroring
One CPU

Mirroring
Two CPU

Memory population rules

There are total of 12 memory slots on the system board.

X

The table contains memory population rule of one CPU with independent mode.

Total DIMMs	Processor 1						Total DIMMs
	6	5	4	3	2	1	
1				3			1
2			4	3			2
3			4	3	2		3
4		5	4	3	2		4
5		5	4	3	2	1	5
6	6	5	4	3	2	1	6

Independent
One CPU

Independent
Two CPU

Rank sparing
One CPU

Rank sparing
Two CPU

Mirroring
One CPU

Mirroring
Two CPU

Memory population rules

X

The table contains memory population rule of two CPUs with independent mode.

Total DIMMs	Processor 2						Processor 1						Total DIMMs
	12	11	10	9	8	7	6	5	4	3	2	1	
2				9						3			2
3				9					4	3			3
4			10	9					4	3			4
5			10	9					4	3	2		5
6			10	9	8				4	3	2		6
7			10	9	8			5	4	3	2		7
8		11	10	9	8			5	4	3	2		8
9		11	10	9	8			5	4	3	2	1	9
10		11	10	9	8	7		5	4	3	2	1	10
11		11	10	9	8	7	6	5	4	3	2	1	11
12	12	11	10	9	8	7	6	5	4	3	2	1	12

Independent
One CPU

Independent
Two CPU

Rank sparing
One CPU

Rank sparing
Two CPU

Mirroring
One CPU

Mirroring
Two CPU

Memory population rules

X

The table contains memory population rule of one CPU with rank sparing mode.

All DIMMs to be installed must be the same type with the same capacity, frequency, voltage, and number of ranks.

Single-rank DIMMs do not support rank sparing mode.

Total DIMMs	Processor 1						Total DIMMs
	6	5	4	3	2	1	
1				3			1
2			4	3			2
3			4	3	2		3
4		5	4	3	2		4
5		5	4	3	2	1	5
6	6	5	4	3	2	1	6

Independent
One CPU

Independent
Two CPU

Rank sparing
One CPU

Rank sparing
Two CPU

Mirroring
One CPU

Mirroring
Two CPU

Memory population rules

X

The table contains memory population rule of two CPUs with rank sparing mode.

Total DIMMs	Processor 2						Processor 1						Total DIMMs
	12	11	10	9	8	7	6	5	4	3	2	1	
2				9						3			2
3				9					4	3			3
4			10	9					4	3			4
5			10	9					4	3	2		5
6			10	9	8				4	3	2		6
7			10	9	8			5	4	3	2		7
8		11	10	9	8			5	4	3	2		8
9		11	10	9	8			5	4	3	2	1	9
10		11	10	9	8	7		5	4	3	2	1	10
11		11	10	9	8	7	6	5	4	3	2	1	11
12	12	11	10	9	8	7	6	5	4	3	2	1	12

Independent
One CPU

Independent
Two CPU

Rank sparing
One CPU

Rank sparing
Two CPU

Mirroring
One CPU

Mirroring
Two CPU

Memory population rules

There are total of 12 memory slots on the system board.

X

The table contains memory population rule of one CPU with mirroring mode.

All DIMMs to be installed must be the same type with the same capacity, frequency, voltage, and rank.

Total DIMMs	Processor 1						Total DIMMs
	6	5	4	3	2	1	
2				3	2		2
3				3	2	1	3
4		5	4	3	2		4
6	6	5	4	3	2	1	6

Independent
One CPU

Independent
Two CPU

Rank sparing
One CPU

Rank sparing
Two CPU

Mirroring
One CPU

Mirroring
Two CPU

Memory population rules

X

The table contains memory population rule of two CPUs with mirroring mode.

Total DIMMs	Processor 2						Processor 1						Total DIMMs
	12	11	10	9	8	7	6	5	4	3	2	1	
4				9	8					3	2		4
5				9	8					3	2	1	5
6				9	8	7				3	2	1	6
7				9	8	7		5	4	3	2		7
8		11	10	9	8			5	4	3	2		8
9				9	8	7	6	5	4	3	2	1	9
10		11	10	9	8		6	5	4	3	2	1	10
12	12	11	10	9	8	7	6	5	4	3	2	1	12

Independent
One CPU

Independent
Two CPU

Rank sparing
One CPU

Rank sparing
Two CPU

Mirroring
One CPU

Mirroring
Two CPU

SR550 chassis matrix

The table contains the SR550 storage and controller configurations. Click arrows to view more information.

		Back-plane	HDD type					M.2 boot module		
Chassis	HDD configuration		3.5-inch SATA	3.5-inch SAS	2.5-inch SATA/SATA SSD with 3.5-inch HDD cage	2.5-inch SAS/SAS SSD with 3.5-inch HDD cage	2.5-inch SAS/SATA SSD	M.2 SSD	Single slot	Dual slot with HW RAID
Eight 3.5-inch LFF	Eight 3.5-inch SATA (SS)		x					x	x	x
	Eight 3.5-inch SAS/SATA (HS)	4x2 3.5-inch	x	x	x	x		x	x	x
12x 3.5-inch LFF	12x 3.5-inch SAS/SATA (HS)	4x3 3.5-inch	x	x	x	x		x	x	x
16x 2.5-inch SFF	No drive / backplane							x	x	x
	Eight 2.5-inch SAS/SATA (HS)	1x8 2.5-inch					x	x	x	x
	16x 2.5-inch SAS/SATA (HS)	Two 1x8 2.5-inch					x	x	x	x

[illegible]

RAID configurations

Click chassis one or chassis two to view the details.

Chassis one: 3.5-inch chassis supports up to 12 bays

Chassis two: 2.5-inch chassis supports up to 16 bays

Chassis one: 3.5-inch chassis supports up to 12 bays

X

Eight 3.5-inch SS SATA

Eight 3.5-inch SAS/SATA

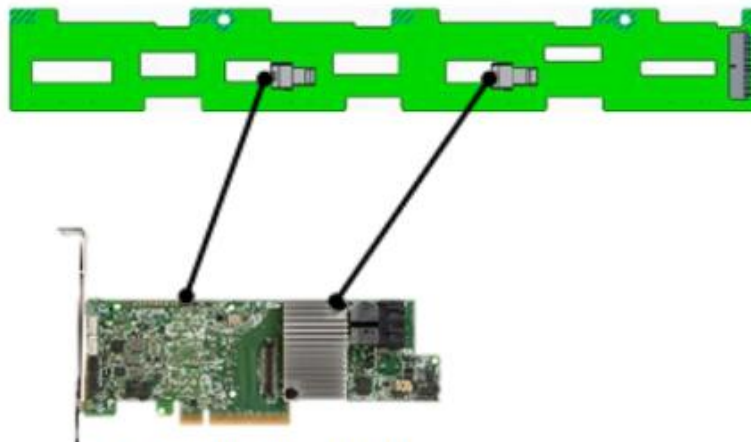
12x 3.5-inch SAS/SATA

Front view

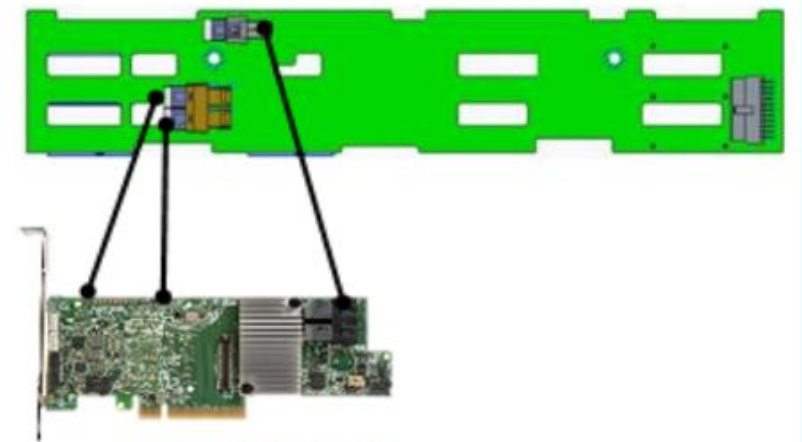


SATA cable from
backplate to onboard
connectors

Rear view



One 8i RAID or HBA



One 16i RAID or HBA

Chassis two: 2.5-inch chassis supports up to 16 bays

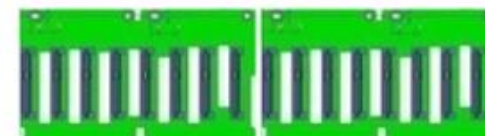
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No backplane, no drive

Eight 2.5-inch SAS/SATA

16x 3.5-inch SAS/SATA

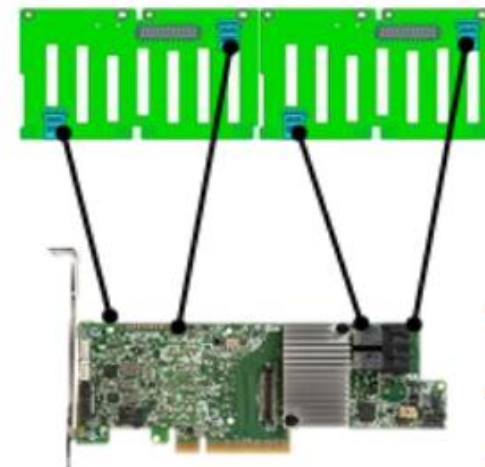
Front view



Rear view



One 8i RAID or HBA



- One 8i RAID or HBA, or
- Two 8i RAID or HBA, or
- One 8i RAID + one 8i HBA

HDD filler population rules

The 3.5-inch chassis only comes with a single 3.5-inch filler. For filler population rules, refer to the chart below.



8x1 2.5-inch filler is used when no backplane is installed.

4x1 2.5-inch filler is used when zero or one HDD is installed.

Single 2.5-inch filler

HDD population rules

SR550 HDD population rules are listed here:

- Mix HDD vendor is supported with Lenovo qualified drives.
- Mix HDD capacity is supported with Lenovo qualified drives.
 - Mixed HDD capacity in one system, but not in one RAID Array.
 - Install lower capacity HDD first.
- Mix 2.5-inch HDD and 3.5-inch HDD into 3.5-inch HDD tray is supported with Lenovo qualified drives.
 - Mix 2.5-inch and 3.5-inch HDD in one system, but not in one RAID Array.
 - Support mix 2.5-inch of SSD/SAS/SATA into 3.5-inch HDD tray.
 - Install 2.5-inch HDD first.
 - Install lower capacity HDD first.
 - HDD type priority: SSD > SAS > SATA
- Mix SSD/SAS/SATA HDD into 2.5-inch HDD tray is supported with Lenovo qualified drives.
 - Mix SSD/SAS/SATA into 2.5-inch HDD tray in one system, but not in one RAID Array.
 - Install lower capacity HDD first.
 - HDD type priority: SSD > SAS > SATA

HDD installation rules

Follow the rules when installing storage components.



SAS/SATA HDD/SSD installation rules:

- 3.5-inch SAS/SATA HDD/SSD can be installed in front bay 0 through 11.
- 2.5-inch SAS/SATA HDD/SSD can be installed in front bay 0 through 15.

M.2 installation rules:

- Install M.2 module onto M.2 slot
- Single M.2 module supports one M.2 media; dual M.2 module supports maximum two M.2 media.
- Install M.2 media on bottom side first for dual M.2 module.